



Analytics & Big Data

Session 8: Big data I

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Learning objectives

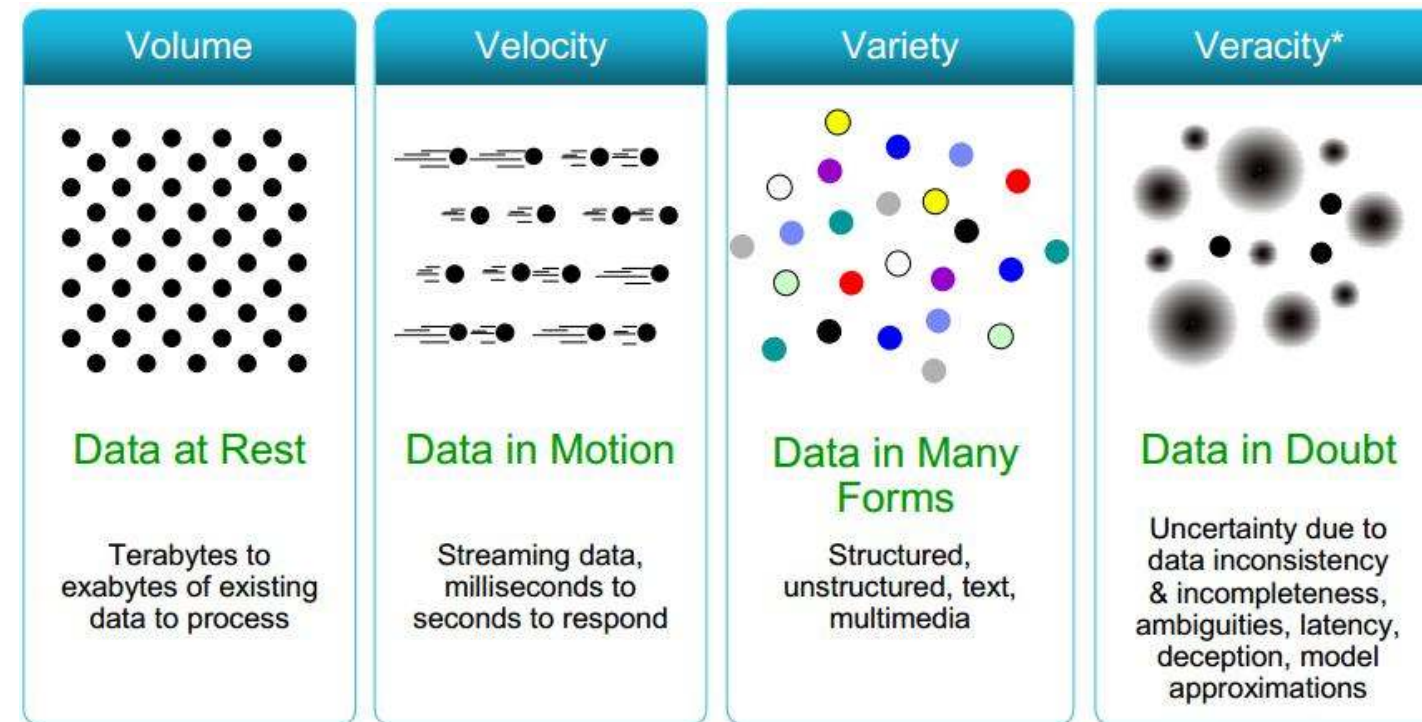
- **Explain** the characteristics of big data and their implications for analytics workflows.
- **Assess** key challenges of big data, including scalability, heterogeneity, and data quality.
- **Outline** distributed data architectures and their role in large-scale analytics.

TODO: Explain how big data environments reshape model development, validation, and deployment. This creates continuity with: Generalization problem, Overfitting, Model evaluation



Big data

4V's of Big Data



— Source: www.cs.kent.edu/~jin/BigData/Lecture1.pptx

Big data changes the managerial problem



Big data matters when the **properties of the data** change what managers can decide, automate, or compete on.

V	Business dilemma
Volume	Too many operational combinations to optimize manually
Velocity	Decisions must be made before the moment passes
Variety	Many data worlds must be integrated
Veracity	Data may look precise but still be false

Volume: UPS route optimization

Managerial dilemma

How do you optimize millions of delivery decisions when every route has thousands of possible variants?

Key data facts

- 250M+ address data points
- 200,000+ route options per typical route
- Parcels, drivers, fleet, road network, customer constraints
- Real-time updates from operations and telematics

Models: vehicle routing · scheduling · ETA prediction · dynamic re-optimization

Outcomes: fewer miles · lower fuel costs · better on-time delivery · operational data advantage



Velocity: Visa fraud detection

Managerial dilemma

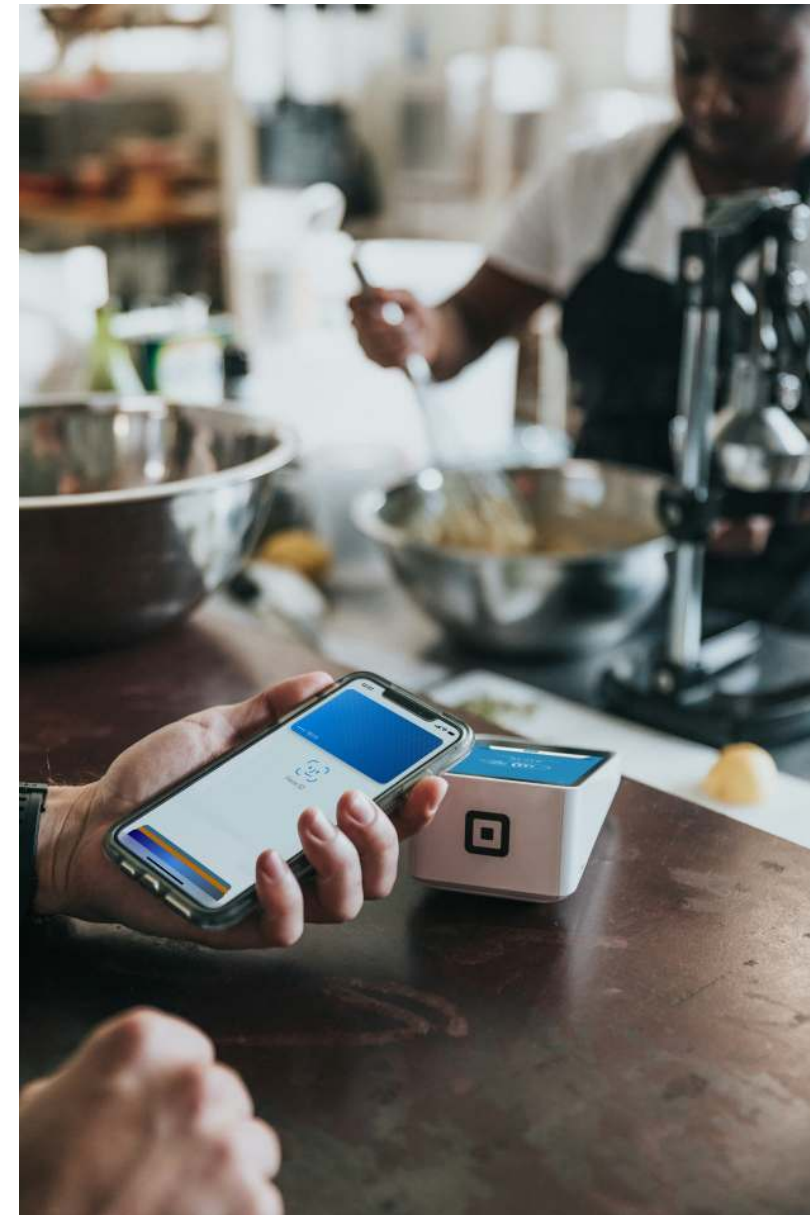
Can fraud be detected **before** the transaction is approved?

Key data facts

- Up to 83,000 transactions / second
- $\approx 7.17\text{B}$ decisions / day at peak capacity
- 500+ data points per transaction
- Fraud is rare, but losses are massive
- Visa reported **\$40B+ fraud blocked in 2023**

Models: real-time risk scoring · anomaly detection · streaming ML · network analysis

Outcomes: approve · decline · challenge · escalate — all within milliseconds



Variety: Walmart ambient IoT supply chain

Managerial dilemma

How do you coordinate inventory when the relevant data comes from many different worlds?

Key data facts

- Target: **90M pallets** by end of 2026
- Active in **500 Walmart locations**
- Potential scale: 4,600 stores + 40+ distribution centers
- Sensor data: location, movement, temperature, humidity, dwell time
- Combined with inventory, supplier, store, and replenishment data

Models: sensor fusion · demand forecasting · replenishment optimization · exception detection

Outcomes: fewer stockouts · fresher products · less waste · tighter supplier coordination



Veracity: Digital advertising fraud

Managerial dilemma

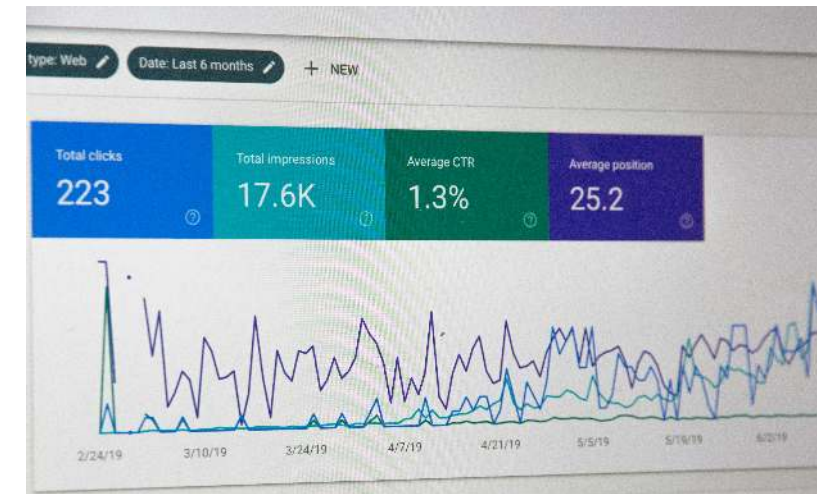
The dashboard says there were clicks and impressions — but were they real?

Key data facts

- Fraudulent impressions, clicks, conversions, or data events
- DoubleVerify measures **8.3T+ media transactions / year**
- Ad fraud losses projected at **\$172B by 2028**
- One fraud operation fell from **2.5B** to **100M** daily bid requests after mitigation

Models: bot detection · invalid traffic scoring · anomaly detection · graph analysis

Outcomes: block fake traffic · clean attribution · reallocate spend · protect ROI



TODO: too simple, but maybe integrate in the “end of the first section” (Key takeaway)

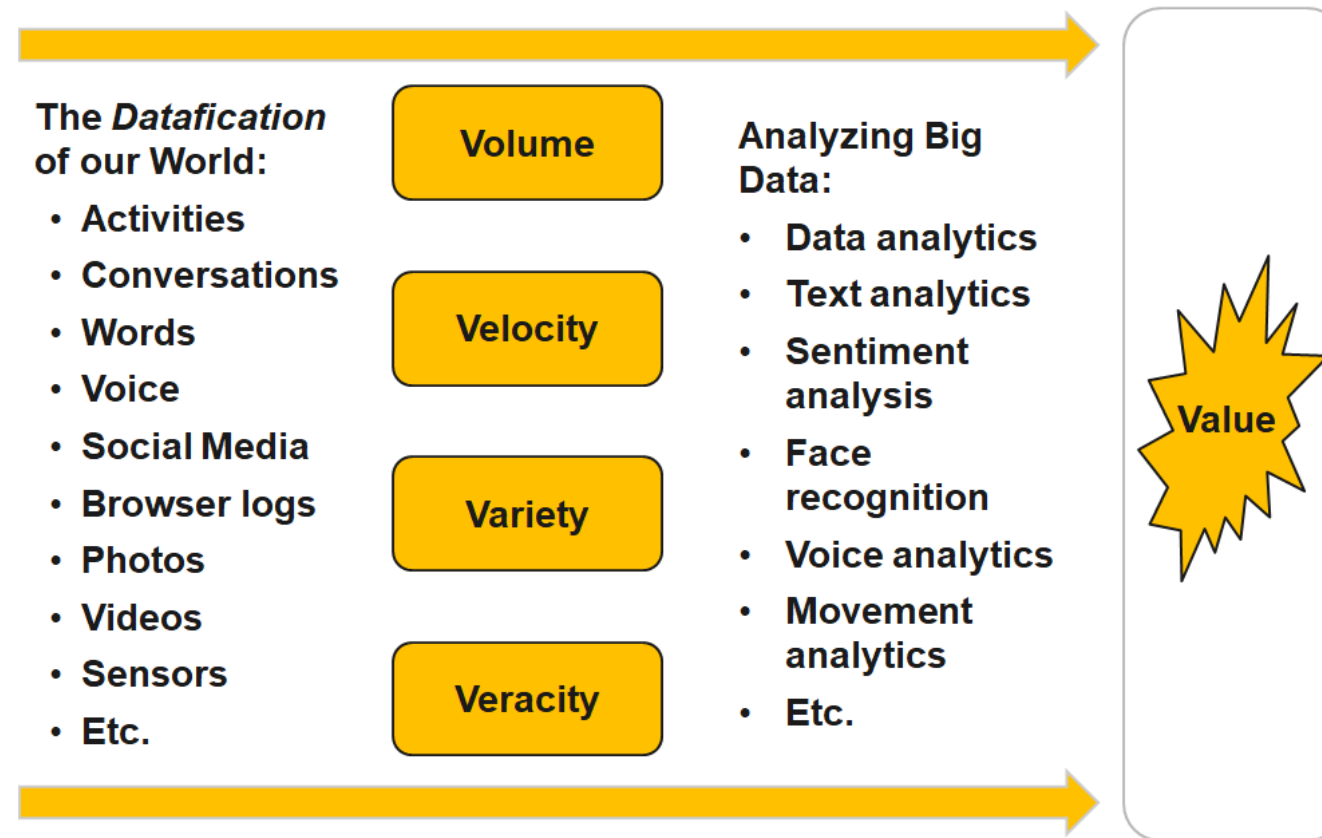


Big data is not one problem.

It creates **four different managerial dilemmas**:

V	Core challenge	Business response
Volume	Too many combinations	Optimize
Velocity	Too little time	Decide instantly
Variety	Too many data worlds	Integrate
Veracity	Too little trust	Validate

Turning Big Data into Value



— Source: http://de.slideshare.net/BernardMarr/140228-big-data-slide-share/3-The_basic_idea_behind_the

Architecture

TODO: data lake vs data warehouse as alternative architectures? clarify the difference!

Limitations of the Data Warehouse Approach

Traditional DWH solutions are designed to provide a single point of truth. Important aspects are:

- Merge and unify data from multiple data sources
- High data quality
- Proper historization of the data
- Data Governance and Compliance

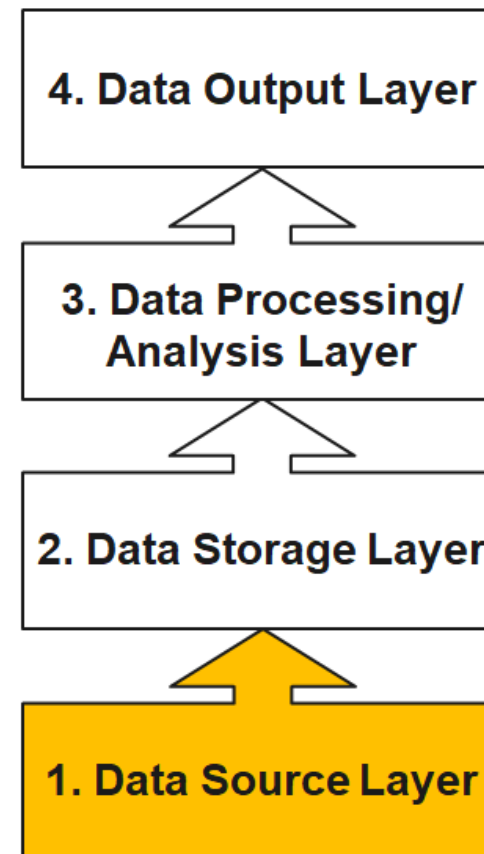
This results in the following problem areas in today's world:

- **Lack of flexibility** due to high effort for changes
- **Time expenditure** due to transfer from the operational sources and the aggregations
- **Past orientation** of data (snapshot of the past); there is an increasing need for ad hoc and real-time analyses
- **Partial knowledge**, as only structured data is stored, with increasing need for social media data, etc.
- **Patchwork**: due to gradual introduction of data warehouses, many isolated solutions exist and are operated separately both technically and methodologically

The Four Layers of Big Data (I)

Data Source Layer

This is where the data arrives at the organization. It includes everything from sales records, customer database, feedback, social media channels, marketing list, email archives etc.



— Source: http://de.slideshare.net/BernardMarr/140228-big-data-slide-share/3-The_basic_idea_behind_the

Identify and Prioritize Data Sources



Key

Worst...

Best

Data Source	Increase Store Traffic	Increase Shopping Bag Revenue	Increase # Corporate Events	Increase Promotional Effectiveness	Improve NPI Effectiveness
Point of Sales Transactions	●	●	●	●	●
Market Baskets	●	●	◐	●	●
Store Demographics (Zip Code)	◐	◐	◐	◐	◐
Local Competitive Stores	◐	◐	◐	◐	◐
Store Manager Demographics	◐	◐	◐	◐	◐
Consumer Comments	◐	◐	◐	◐	◐
Social Media	◐	◐	◐	◐	◐
Weather	◐	◐	◐	◐	◐
Local Events	●	◐	◐	◐	◐
Traffic	◐	◐	◐	◐	◐
Zillow	◐	◐	◐	◐	◐

Business value of potential data sources

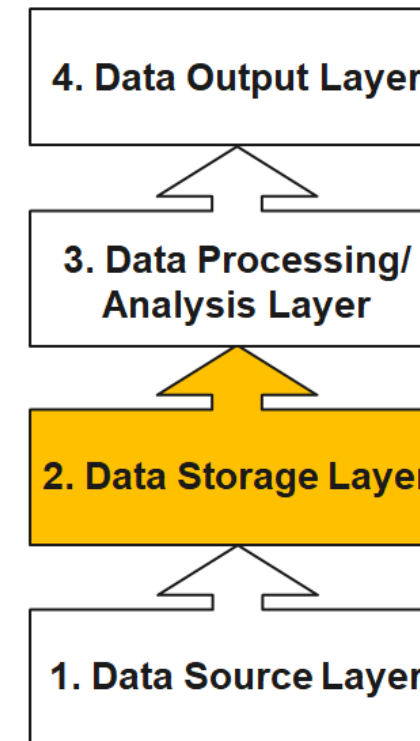
— Source: Schmarzo (2016), p. 49

The Four Layers of Big Data (II)



Data Storage Layer

This is where Big Data is stored, once it is gathered from the sources. As the volume of data stored has started to explode, new database technologies have been developed — such as NoSQL database systems.



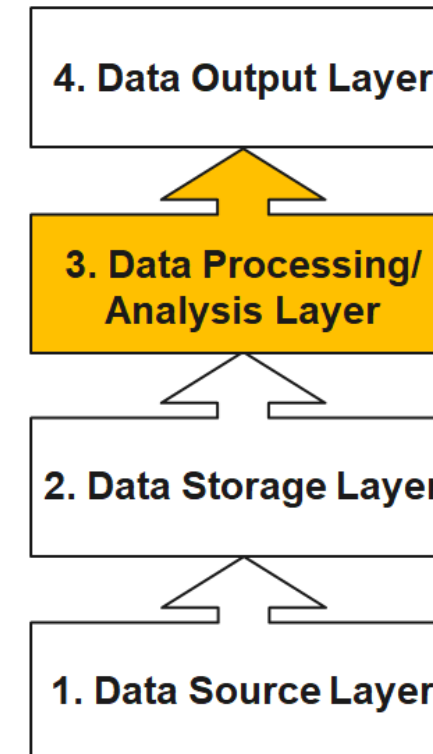
— Source: http://de.slideshare.net/BernardMarr/140228-big-data-slide-share/3-The_basic_idea_behind_the

The Four Layers of Big Data (III)



Data Processing/Analysis Layer

When you want to use the data you have stored to find out something useful, you will need to process and analyze it.



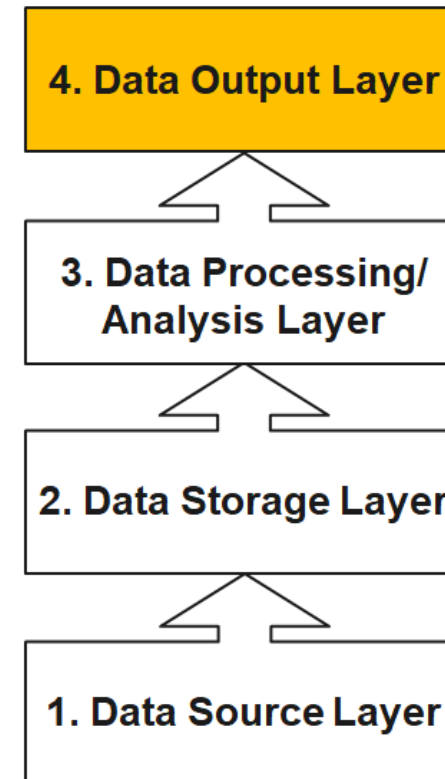
— Source: http://de.slideshare.net/BernardMarr/140228-big-data-slide-share/3-The_basic_idea_behind_the

The Four Layers of Big Data (IV)



Data Output Layer

This is how the insights gleaned through the analysis is passed on to the people who can take action to benefit from them.



— Source: http://de.slideshare.net/BernardMarr/140228-big-data-slide-share/3-The_basic_idea_behind_the

From Data Warehouse to Data Lake

Instead of recording millions of transactions, today's organizations are recording billions of interactions. Companies are capturing more and more data that can open business opportunities and unlock new sources of value for organizations.

Companies are not able to store this data in data warehouses because it is of high volume, mostly raw and often not structured. As consequence, data lakes have emerged as an alternative approach. The intent is to capture enterprise data and load it in its raw form into a centralized, large, and inexpensive storage system.

In shifting from data warehouses to data lakes, it became important to decouple data movement from data transformation. Data movement (the “E” and “L” of ETL) is an operational task. Data transformation (the “T” of ETL) is a content-based, analytic-facing task that requires an understanding both of the data and how it's to be used.

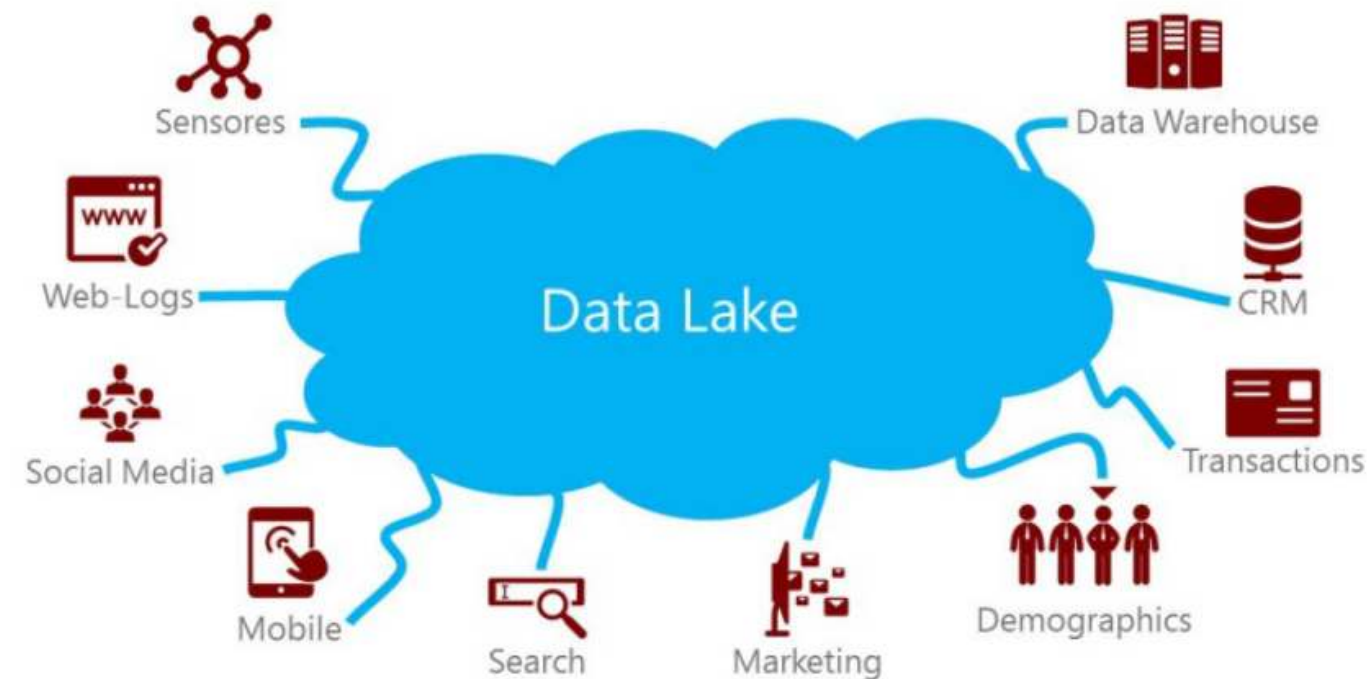
A clean separation between data movement and data transformation has the benefits of less friction because the instance loading the data isn't responsible for transforming it.

— Source: Trifacta: EOL for

The Data Lake

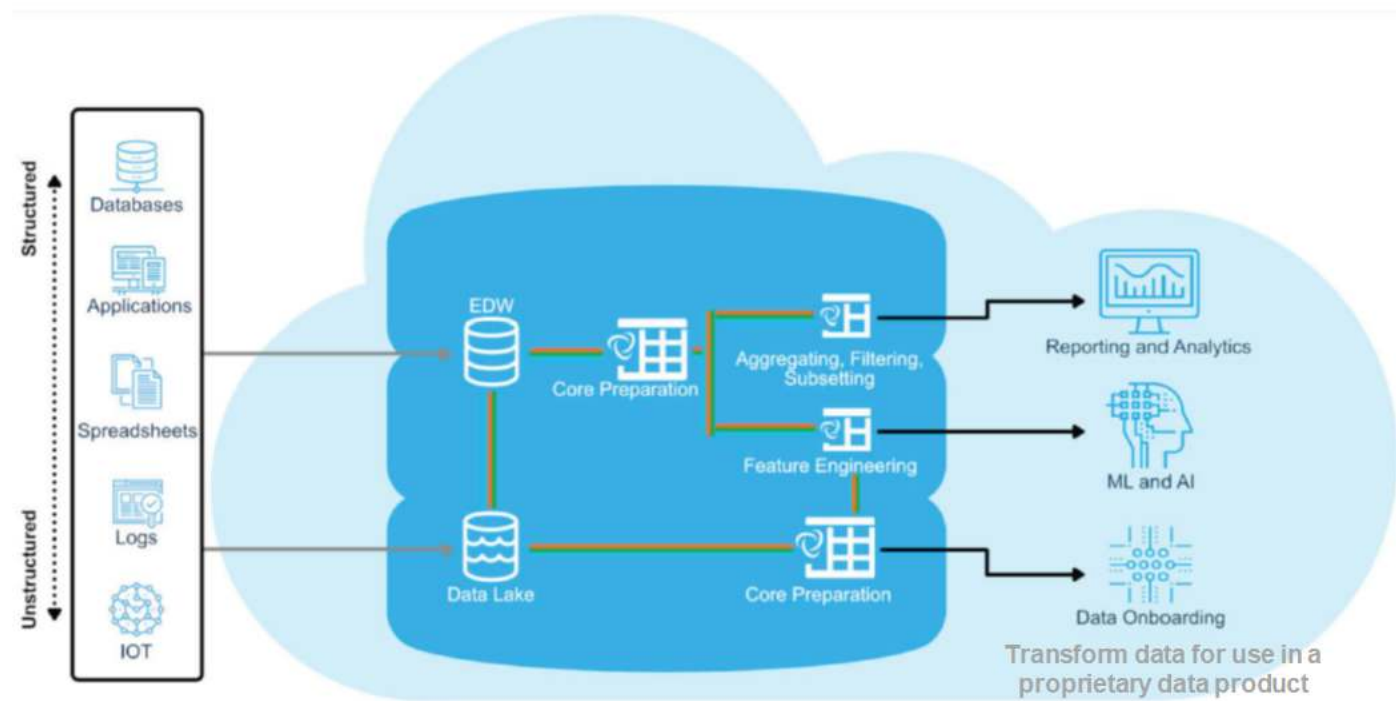


A data lake is a method of storing data within a system in its natural format, that facilitates the collocation of data in various schemata and structural forms. The idea of data lake is to have a single store of all data in the enterprise ranging from raw data to transformed data which is used for various tasks including reporting, visualization, analytics and machine learning.



— Source: https://www.pmone.com/fileadmin/user_upload/pics/other/Data_lake.jpg

Modern Data Analytics Architecture



Source: Trifacta: EOL for ETL?

Logical Data Warehouse

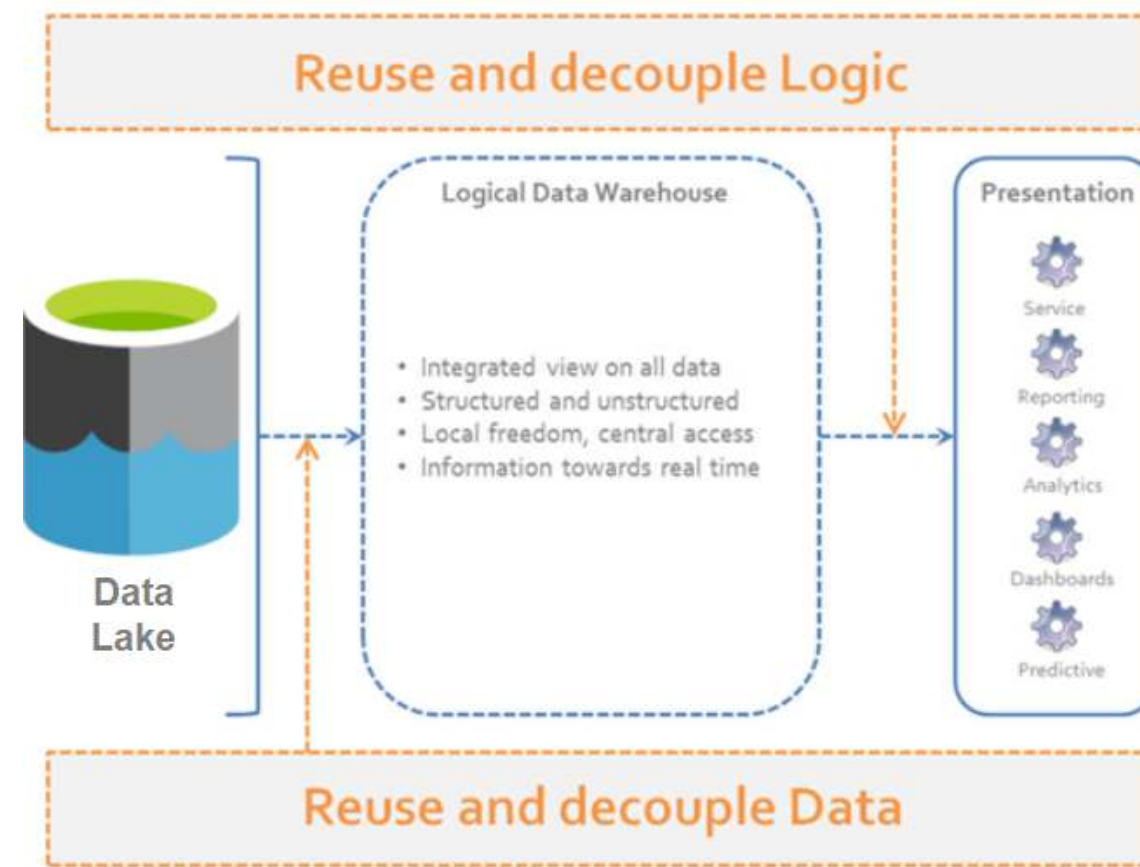


A “logical data warehouse” provides analytical company data without first physically moving it to a physical data warehouse.

As in a classic data warehouse, uniform views are provided for analysis purposes.

While the data in the classic data warehouse comes from a “well-defined” physically uniform database, the “logical data warehouse” pulls data together from the data lake at the time of the query.

Aggregation is done just in time. Thus, the schema of the data warehouse is just virtual.



— Source: <http://www.datavirtualizationblog.com/emergence-logical-data-warehouse/>

Text analytics

TODO: structure: see “Text representation pipeline” note (obsidian)

<https://github.com/iam-salma/NLP-Bootcamp-with-python>

Paradigm 1: Sparse Representations (Classical NLP)

- Core challenge: *How do we turn text into analyzable data?*
- Basic pipeline:
 - preprocessing (tokenization, cleaning)
 - feature engineering (Bag-of-Words, TF-IDF)
- Output: **sparse feature vectors** → ML models
- Use case: classification (e.g., sentiment analysis)
- Key idea: *text* → *structured matrix* → *prediction*

Paradigm 2: Dense Representations (Embeddings)

- Motivation: capture **semantic meaning**
- Introduce embeddings (e.g., BERT)
- Output: **dense vectors** → ML / clustering / similarity
- Use cases:
 - semantic search
 - clustering documents
 - regression/classification
- Key shift: *better features, same analytics logic*

Paradigm 3: AI-Assisted Analytics (LLMs & Systems)

- Embeddings as foundation for systems:
 - retrieval (finding relevant documents)
 - similarity-based analytics
- Extend to LLM-based workflows (e.g., GPT-4):
 - direct classification via prompting
 - information extraction
 - summarization
- Concept: **Retrieval-Augmented Generation (RAG)**
 - embeddings → retrieve relevant data
 - LLM → analyze / generate output
- Key shift: *model integrates feature extraction + prediction*

5. Comparison & Analytical Perspective

- Three ways to turn text into insights:

Approach	Representation	Role in Analytics
Sparse	TF-IDF	baseline, interpretable
Dense	embeddings	semantic features
LLM-based	implicit	flexible, system-level analytics

- Emphasize: **choice depends on task, cost, and requirements**

6. Takeaway: The Future of Text Analytics

- Text is no longer “unstructured” → it’s **transformable data**
- Evolution:
 - manual features → learned representations → AI-assisted analysis
- Role of analytics:
 - selecting representations
 - designing pipelines/systems
 - evaluating outcomes

TODO Deployment



Cover ML Ops pipelines, scaling (parallel architectures)

Survey: Session 8



<https://forms.gle/hNuAD8UZvC8KYo4a8>

Note: Responses may be analyzed and published in anonymized form.

Please complete the survey before you leave today — thank you 🙏

References



Schmarzo, B. (2016). *Big data MBA*. Wiley. <https://doi.org/10.1002/9781119238881>